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Bearing Damage, ISO Classification, and Terminology

Presented at the bearing damage course, WE 204
Presented by xxx
2025-05-29

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ISO 15243:2004

*Rolling bearings – Damage and failures -
Terms, characteristics, and causes*

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How to acquire the ISO 15243 standard...

www.iso.org



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How to acquire the ISO 15243 standard

ISO 15243:2004

Rolling bearings -- Damage and failures -- Terms, characteristics and causes

Edition:
Number of pages:
Technical committee / subcommittee:
ICS:
Status:
Current stage:
Stage date:
Revision information:

1 (Monolingual)
41
[TC 4; ISO Standards](#)
[21.100.20](#)
✔ Published standard
[60.60](#)
2004-02-11
None

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	Size	Price
 ISO 15243:2004 PDF version (en)	16980 KB	CHF 138,00
 ISO 15243:2004 PDF version (fr)	17049 KB	CHF 138,00
 ISO 15243:2004 paper version (en)		CHF 138,00
 ISO 15243:2004 paper version (fr)		CHF 138,00

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ISO standard 15243

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ISO 15243:2004 – scope

The classification is based on three main points:

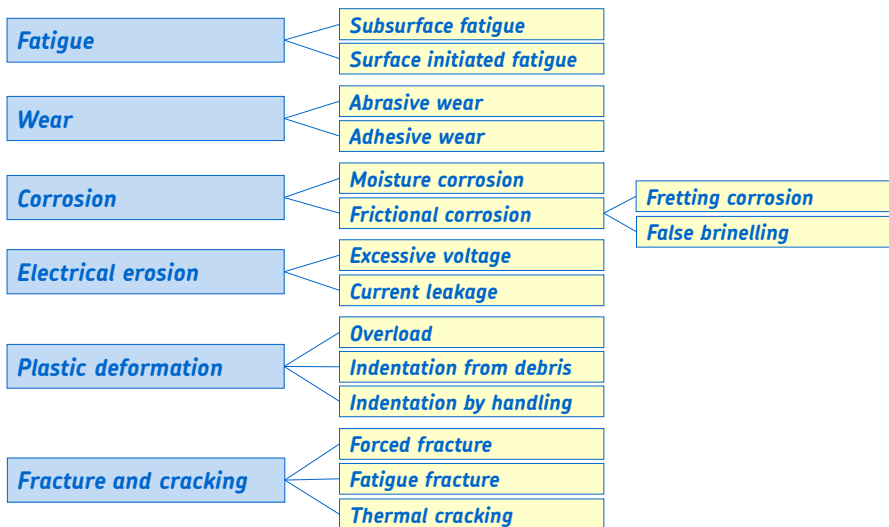
- 1. Damage and changes in appearance DURING SERVICE.** It means that damage prior to service is not classified in the ISO standard (service = as soon as the bearing leaves the factory).
- 2. CHARACTERISTIC FORMS of change in appearance, which can be attributed to particular causes with a high degree of certainty (extreme, unlikely cases, therefore not classified).**
- 3. Classification by VISIBLE FEATURES.**

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Failure modes: ISO classification



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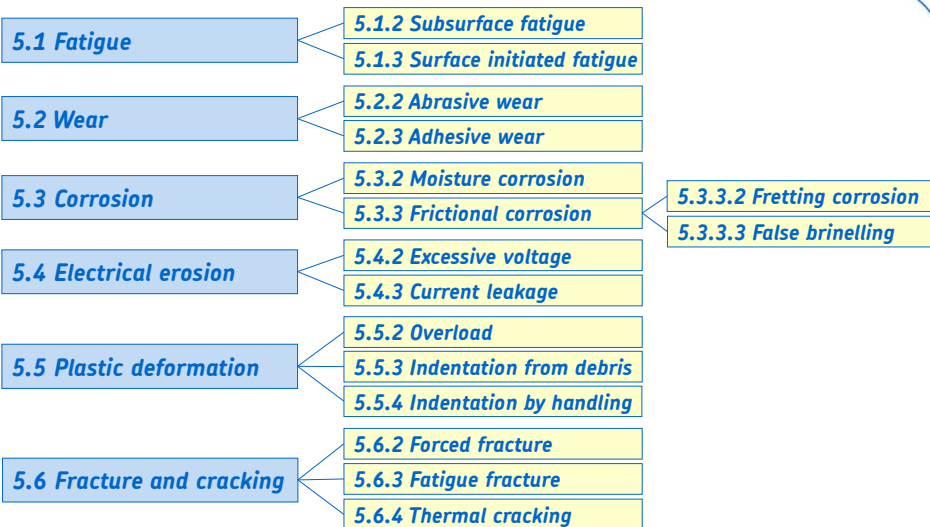
ISO classification (ISO 15243)

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Failure modes: ISO classification



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The ISO classification system

- *The ISO classification system is made up of six main modes and then further divided into sub-modes*
- *In total, 15 categories can be observed*
- *These categories will be covered, one by one, indicating the typical features; a number of typical examples are shown*
- *There are some other reasons for bearing damage, such as design problems, etc. These are not classified in the ISO standard*

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5.1

Fatigue

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5.1.1 general definition

- *The change in the structure, which is caused by the repeated stresses developed in the contacts between the rolling elements and the raceways*
- *Fatigue is manifested visibly as a flaking of particles from the surface*
- *Fatigue can be divided into:*

5.1.2 Subsurface initiated fatigue

5.1.3 Surface initiated fatigue

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5.1.2

Subsurface initiated fatigue

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5.1.2 subsurface initiated fatigue...

5.1 Fatigue

5.2 Wear

5.3 Corrosion

5.4 Electrical erosion

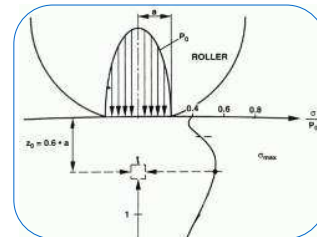
5.5 Plastic deformation

5.6 Fracture and cracking

5.1.2 Subsurface fatigue

5.1.3 Surface initiated fatigue

- Repeated stress changes
- Material structural changes
- Micro-cracks under the surface
- Crack propagation
- Spalling (also known as flaking & peeling)



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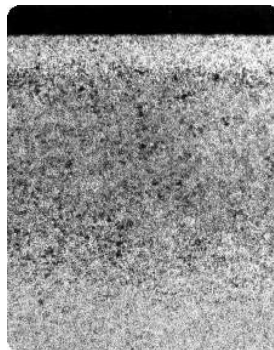
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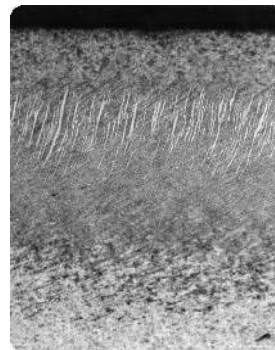
5.1.2 subsurface initiated fatigue...



10 million



63 million



630 million

Number of Total Revolutions

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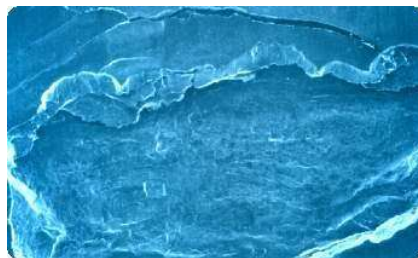
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5.1.2 subsurface initiated fatigue...



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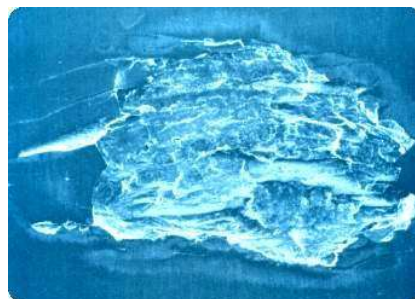
5.1.2 subsurface initiated fatigue...



Normal fatigue

regular or smooth bottom
Very clean steel

rough bottom surface
Not clean steel



Premature spalling

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5.1.2 subsurface initiated fatigue...



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5.1.2 subsurface initiated fatigue...



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5.1.2 subsurface initiated fatigue

5.1 Fatigue

5.1.2 Subsurface fatigue

How to avoid?

- *Make sure you have the right bearing for the application*
- *Use only the highest quality bearings*
- *SKF Explorer bearings provide extended service life*
- *Make sure that components around the bearing are in good condition*
- *Mount correctly*

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5.1.3

Surface initiated fatigue

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5.1.3 surface initiated fatigue...

5.1 Fatigue

5.2 Wear

5.3 Corrosion

5.4 Electrical erosion

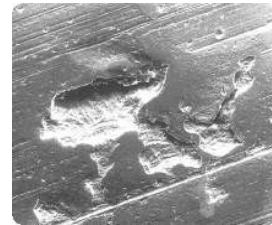
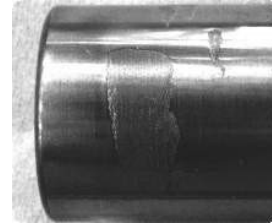
5.5 Plastic deformation

5.6 Fracture and cracking

5.1.2 Subsurface fatigue

5.1.3 Surface initiated fatigue

- Reduced lubrication regime
- Surface distress
- Sliding motion
- Burnishing, glazing
- Asperity micro-cracks
- Asperity micro-spalls



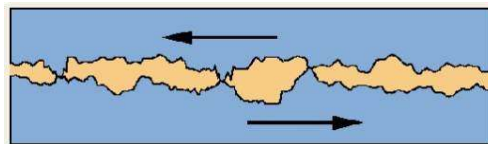
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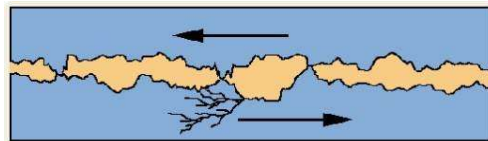
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5.1.3 surface initiated fatigue...

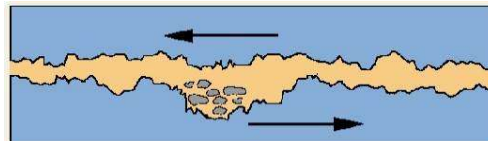
General surface distress happen when



asperities coming in direct contact under mixed or boundary lubrication conditions



When the loading and the frictional forces rise to a given magnitude, small cracks form in the surface



these cracks can in turn develop into micro-spalls

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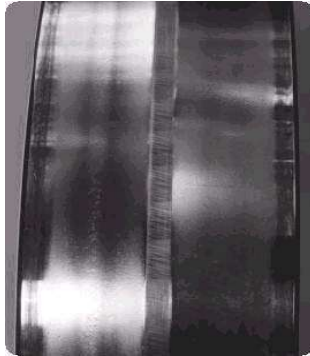
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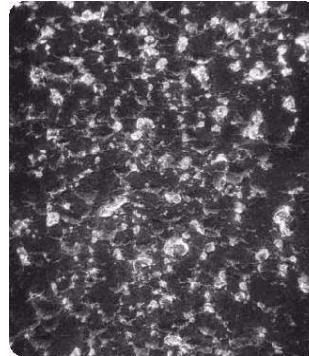
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5.1.3 surface initiated fatigue...



micro-spalls are not visible to the naked eye, because they are only a few microns in size
The surface just looks dull and grey



under a microscope a number of cracks and spalls can be detected

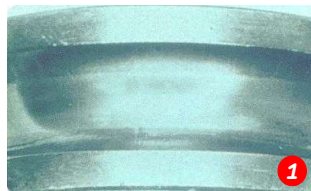
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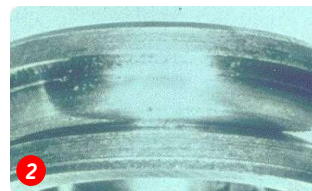
5.1.3 surface initiated fatigue...

Fine roughening or waviness from inadequate lubrication

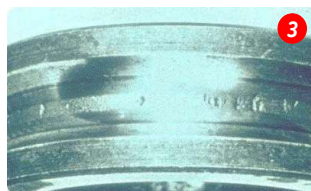


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Surface distress and small cracks develop, and then micro-spalling occurs



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Debris are over-rolled, real (full size) surface spalling develops



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Subsequently whole raceway is spalled

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5.1.3 surface initiated fatigue

5.1 Fatigue

5.1.3 Surface initiated fatigue

How to avoid?

- *Make sure you have the right bearing for the application*
- *Make sure you operate with adequate lubrication*
 - *The right lubricant*
 - *The right quantity*
 - *At the right time*
- *Check the quality of the lubricant at regular intervals*

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5.2

Wear

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5.2.1 general definition

- *Wear is the progressive removal of material resulting from the interaction of the asperities of two sliding or rolling/sliding contacting surfaces during service*
- *Wear can be divided into:*

5.2.2 Abrasive wear

5.2.3 Adhesive wear

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5.2.2

Abrasive wear

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5.2.2 abrasive wear...

5.1 Fatigue

5.2 Wear

5.3 Corrosion

5.4 Electrical
erosion

5.5 Plastic
deformation

5.6 Fracture and
cracking

5.2.2 Abrasive wear

5.2.3 Adhesive wear

- Progressive removal of material
- Inadequate lubrication
- Ingress of dirt particles
- Dull surfaces (mostly)
- (Sometimes very shiny)
- Accelerating process

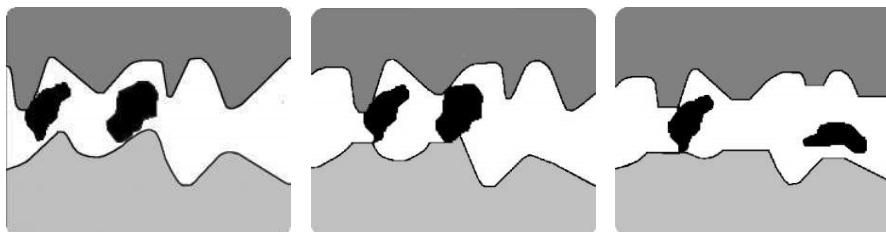


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5.2.2 abrasive wear...



Start

After a number of
cycles

Later

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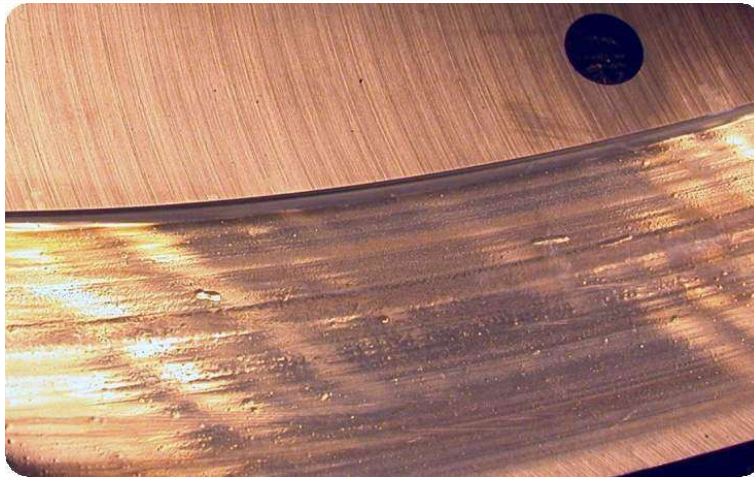
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5.2.2 abrasive wear...



Abrasive wear in a spherical roller bearing outer ring (one raceway).

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5.2.2 abrasive wear...



Abrasive wear on one roller of a spherical roller bearing

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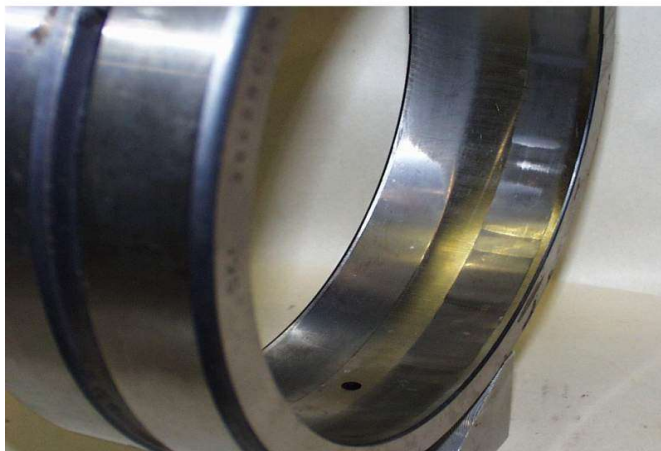
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5.2.2 abrasive wear...



Heavy wear in an outer ring of a spherical roller bearing. The waviness pattern is due to vibration.

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5.2.2 abrasive wear



Spherical roller bearing, extremely heavy wear.

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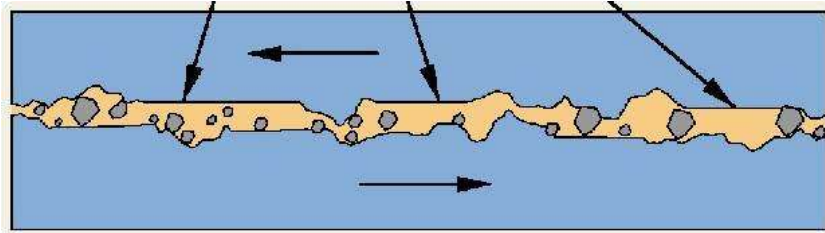
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5.2.2 abrasive wear – polishing effect

Flat surfaces due to polishing and plastic deformation of the surface asperities



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5.2.2 abrasive wear...



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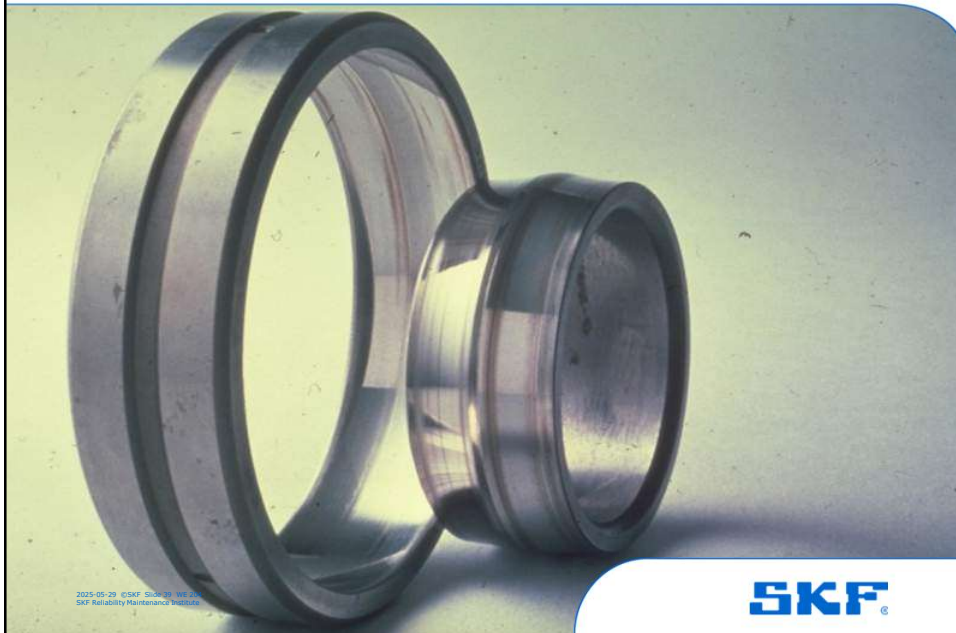
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5.2.2 abrasive wear...



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5.2.2 abrasive wear...



Rings and rolling elements are hardened to around 60 HRC.
A sheet metal cage most of the time is made from soft (unhardened) steel. If lubrication fails, the cage might be the first component to collapse.

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5.2.2 abrasive wear...



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5.2.2 abrasive wear

5.2 Wear

5.2.2 Abrasive wear

How to avoid?

- *Make sure you have an adequate sealing arrangement*
- *Make sure you operate with adequate lubrication*
- *Check the quality of the lubricant at regular intervals*

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5.2.3

Adhesive wear

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5.2.3 adhesive wear...

5.1 Fatigue

5.2 Wear

5.3 Corrosion

5.4 Electrical
erosion

5.5 Plastic
deformation

5.6 Fracture and
cracking

5.2.2 Abrasive wear

5.2.3 Adhesive wear

- Accelerations
- Smearing/skidding/galling
- Material transfer/friction heat
- Tempering/re-hardening with stress concentrations and cracking or flaking
- Low load/high speed



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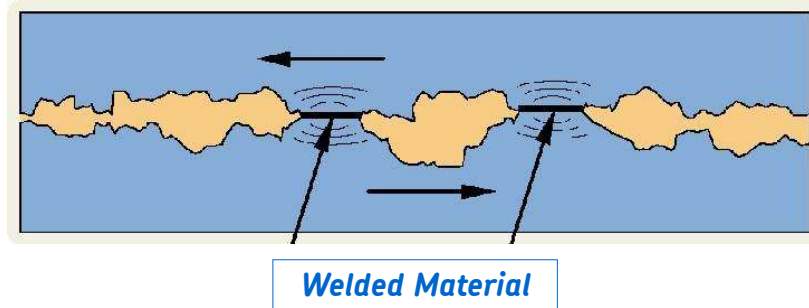
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5.2.3 adhesive wear...

Adhesive wear, just like most other lubrication-related damage, occurs between two mating surfaces



The rollers are accelerated as they enter the loaded zone in bearings operating at high speeds. The heat generation in these sliding contacts may be so great as to cause the oil film to collapse. The two surfaces melt together at the points of metal-to-metal contact. This welding process causes material to be transferred from one surface to the other.

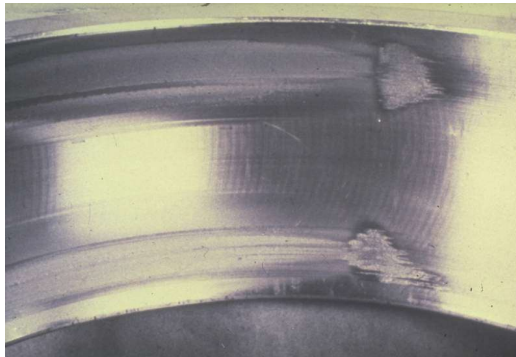
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5.2.3 adhesive wear...

the rollers are accelerated as they enter the loaded zone in bearings operating at high speeds. The heat generation in these sliding contacts may be so great as to cause the oil film to collapse. The two surfaces melt together at the points of metal-to-metal contact



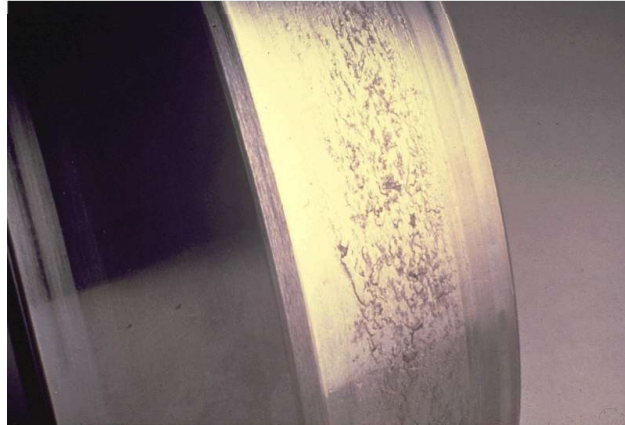
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5.2.3 adhesive wear...



Smearing can also occur between rolling elements and raceways when the load is too light in relation to the speed of rotation.

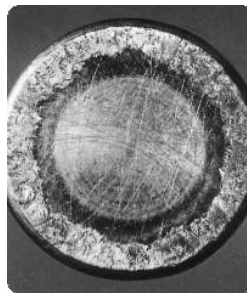
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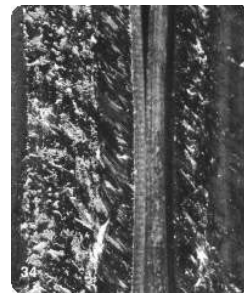
5.2.3 adhesive wear...



***Roller of a spherical
roller bearing
(100 x magnification)***



***End face of a roller
of a cylindrical
roller bearing***



***Guide flange of a
cylindrical roller
bearing***

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5.2.3 adhesive wear

5.2 Wear

5.2.3 Adhesive wear

How to avoid?

- *Make sure the bearing is adequately loaded*
- *Improve the lubricant film thickness*
- *Use lubricants with effective EP additives*
- *For difficult operating conditions (very light loads), using specific bearings can be beneficial:*
 - *Hybrid bearings*
 - *NoWear coating*
- *Downsizing might be of help*

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5.3

Corrosion

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5.3.1 general definition

- *Corrosion is a chemical reaction on metal surfaces*
- *Corrosion can be divided into:*

5.3.2 Moisture corrosion

5.3.3 Frictional corrosion

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5.3.2

Moisture corrosion

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5.3.2 moisture corrosion...

5.1 Fatigue

5.2 Wear

5.3 Corrosion

5.4 Electrical erosion

5.5 Plastic deformation

5.6 Fracture and cracking

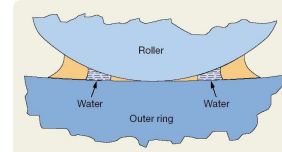
5.3.2 Moisture corrosion

5.3.3 Frictional corrosion

5.3.3.2 Fretting corrosion

5.3.3.3 False brinelling

- Oxidation/rust (corrosion)
- Chemical reaction
- Corrosion pits/flaking
- Etching (water/oil mixture)

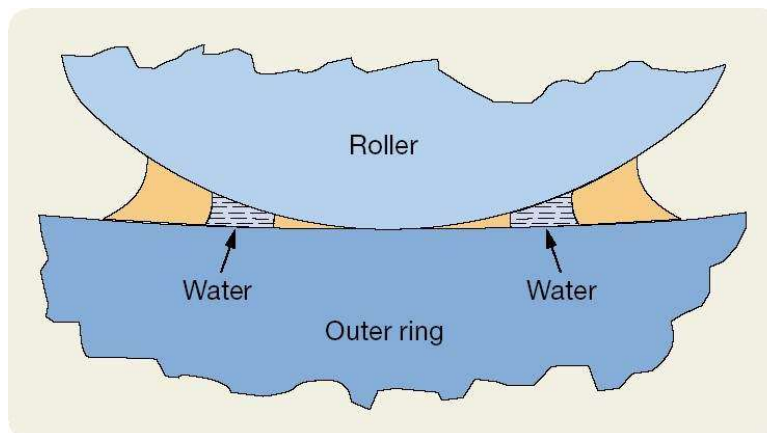


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5.3.2 moisture corrosion...



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5.3.2 moisture corrosion...



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5.3.2 moisture corrosion...



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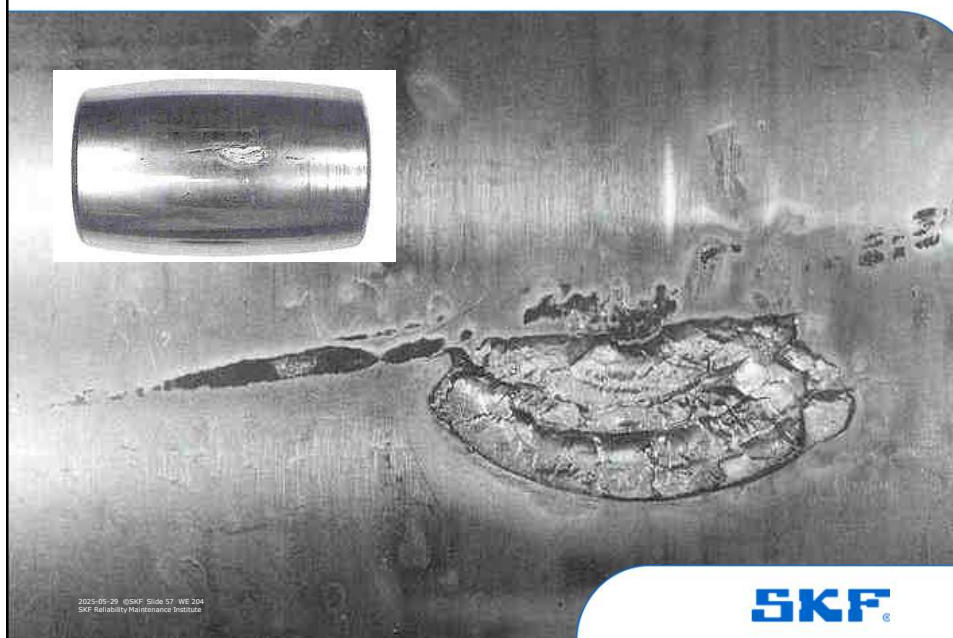
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5.3.2 moisture corrosion...



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5.3.2 moisture corrosion...



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5.3.2 moisture corrosion...



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5.3.2 moisture corrosion

5.3 Corrosion

5.3.2 Moisture corrosion

How to avoid?

- *Adequately protected bearing arrangement (sealing)*
- *Sealed bearings might be beneficial in difficult environments*
- *Make sure you operate with adequate lubrication*
 - *The right lubricant (water repellent)*
 - *The right quantity*
 - *At the right time*

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5.3.3

Frictional corrosion

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5.3.3.1 general definition

- *Frictional corrosion is a chemical reaction activated by relative micro-movements between mating surfaces under certain friction conditions*
- *These micro-movements lead to oxidation of the surfaces and material, becoming visible as powdery rust and/or loss of material from one or both mating surfaces*

Frictional corrosion can be divided into:

- *5.3.3.2 Fretting corrosion*
- *5.3.3.3 False brinelling*

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5.3.3.2

Fretting corrosion

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5.3.3.2 fretting corrosion...

5.1 Fatigue

5.2 Wear

5.3 Corrosion

5.4 Electrical
erosion

5.5 Plastic
deformation

5.6 Fracture and
cracking

5.3.2 Moisture corrosion

5.3.3 Frictional corrosion

5.3.3.2 Fretting corrosion

5.3.3.3 False brinelling

- Micro-movement between mating parts
- Oxidation of asperities
- Powdery rust/loss of material
- Occurs in interfaces transmitting loads



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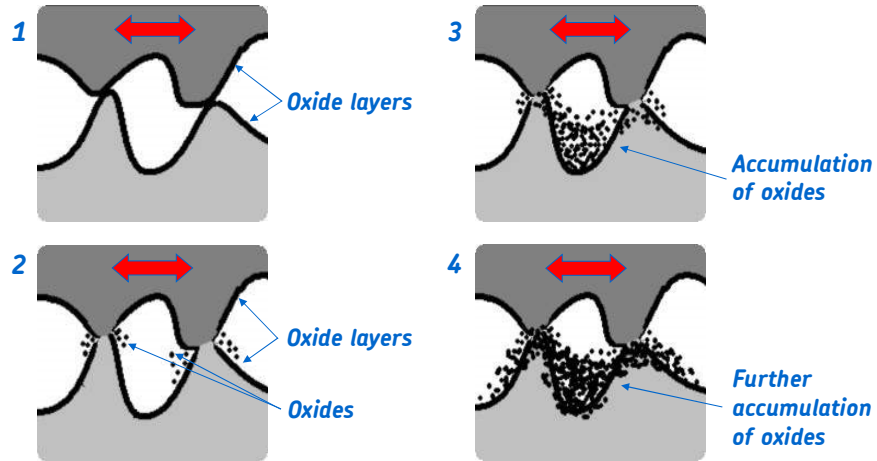
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5.3.3.2 fretting corrosion...

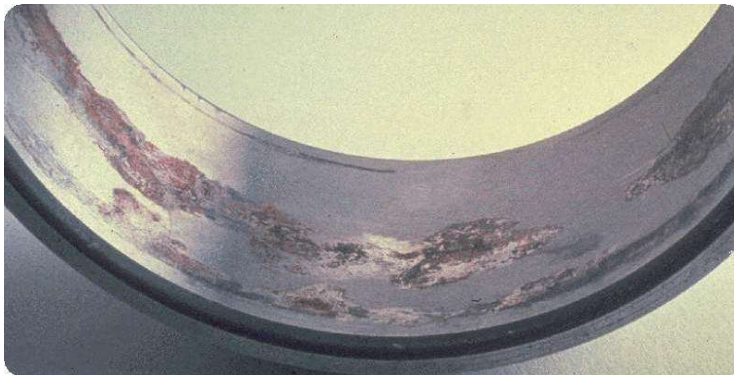


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5.3.3.2 fretting corrosion...



Fretting corrosion because of an imperfect shaft surface (machining!), it can also be due to shaft bending

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5.3.3.2 fretting corrosion...



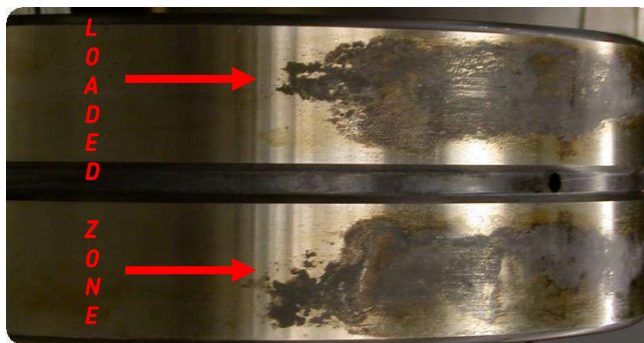
Appearance: Areas of rust on the outside surface of the outer ring or in the bore of the inner ring.
The raceway path pattern could be heavily marked at corresponding positions.
Careful: Fretting corrosion comes as Fe_2O_3 (white colour) or Fe_3O_4 (red colour).

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5.3.3.2 fretting corrosion...



“Heavy” fretting corrosion can often be observed in heavily loaded applications.
Over time, improper contact will result in advanced fretting corrosion and cracking!

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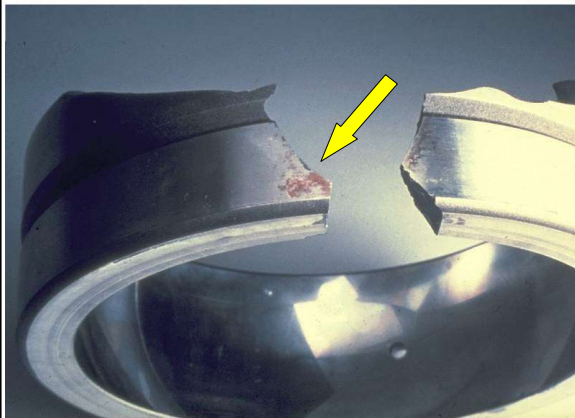
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5.3.3.2 fretting corrosion...



Corroded areas also act as fracture notches.
The outer ring was not well supported in the housing seat.
This gave rise to fretting corrosion, high stresses, and outer ring cracking.

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5.3.3.2 fretting corrosion

5.3 Corrosion

5.3.3 Frictional corrosion

5.3.3.2 Fretting corrosion

How to avoid?

- *Make sure the bearing seats are adequately machined*
- *The seats must be round and appropriate fit used*
- *For a loose fit, coat the seat with an anti-fret paste, e.g., SKF LGAF 3E*

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5.3.3.3

False brinelling

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5.3.3.3 false brinelling...

5.1 Fatigue

5.2 Wear

5.3 Corrosion

5.4 Electrical
erosion

5.5 Plastic
deformation

5.6 Fracture and
cracking

5.3.2 Moisture corrosion

5.3.3 Frictional corrosion

5.3.3.2 Fretting corrosion

5.3.3.3 False brinelling

- Rolling element/raceway
- Micro movements/elastic deformations
- Vibrations
- Corrosion/wear, shiny or reddish depressions
- If stationary: at rolling element pitch
- If rotating: parallel flutes



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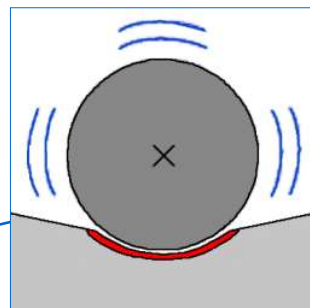


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5.3.3.3 false brinelling...



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5.3.3.3 false brinelling...



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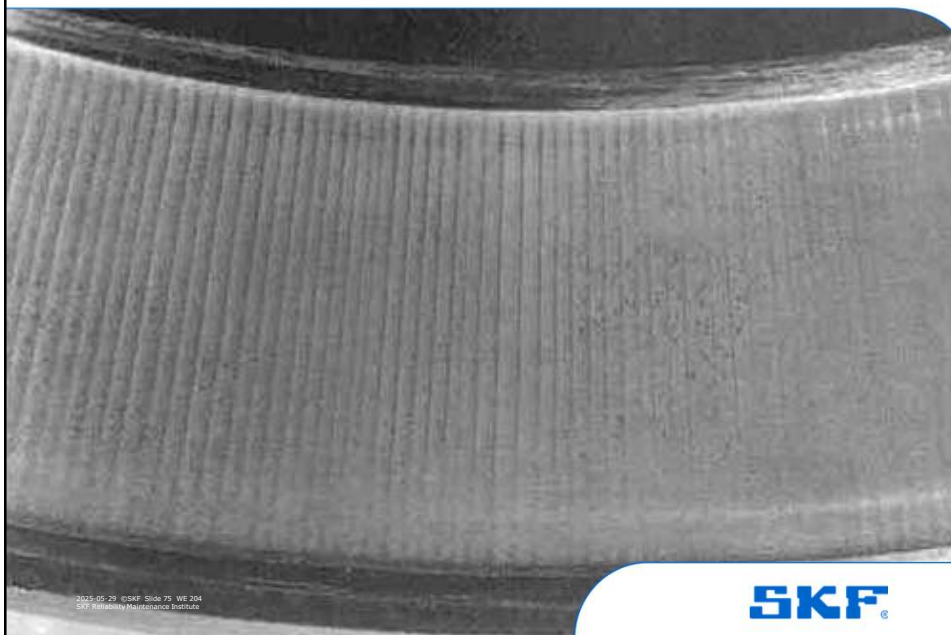
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5.3.3.3 false brinelling...



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5.3.3.3 false brinelling...



Spherical roller bearing shows an extreme case of false brinelling, where the vibrations were (very) heavy over a long period of time

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5.3.3.3 false brinelling

5.3 Corrosion

5.3.3 Frictional corrosion

5.3.3.3 False brinelling

How to avoid?

- *Make sure the vibration levels are within limits*
- *Apply appropriate lubricant*
- *Rotate standby machines at regular intervals*
- *Use adequate bearings for vibratory applications*

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5.4

Electrical erosion

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5.4.1 general definition

- *Electrical erosion is the removal of material from the contact surfaces caused by the passage of electric current*
- *Electrical erosion can be divided into*

5.4.2 Excessive voltage

5.4.3 Current leakage

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5.4.2

Excessive voltage

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5.4.2 excessive voltage...

5.1 Fatigue

5.2 Wear

5.3 Corrosion

5.4 Electrical erosion

5.4.2 Excessive voltage

5.4.3 Current leakage

5.5 Plastic deformation

5.6 Fracture and cracking

- High current: sparking
- Localized heating in very short interval: melting/welding
- Craters up to 0.100 mm



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5.4.2 excessive voltage...

(Small) Craters



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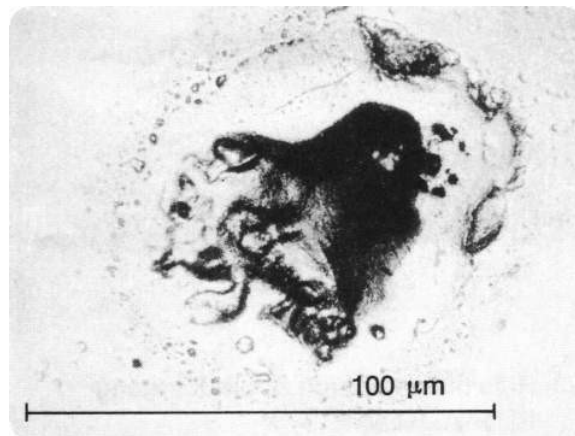
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5.4.2 excessive voltage...

Enlargement of a crater.



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5.4.2 excessive voltage...



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5.4.2 excessive voltage

5.4 Electrical erosion

5.4.2 Excessive voltage

How to avoid?

- *Make sure the connection to earth is done properly*
- *Use insulated bearings*
 - *INSOCOAT*
 - *Hybrid bearings*

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5.4.3

Current leakage

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5.4.3 current leakage...

5.1 Fatigue

5.2 Wear

5.3 Corrosion

5.4 Electrical erosion

5.5 Plastic deformation

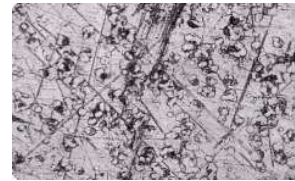
5.6 Fracture and cracking

- Low current intensity
- Shallow craters closely positioned

5.4.2 Excessive voltage

5.4.3 Current leakage

- Development of flutes on raceways and rolling elements, parallel to the rolling axis
- Dark grey discoloration
- Flux asymmetries
- Unsymmetrical cabling
- High frequency currents (frequency converters)

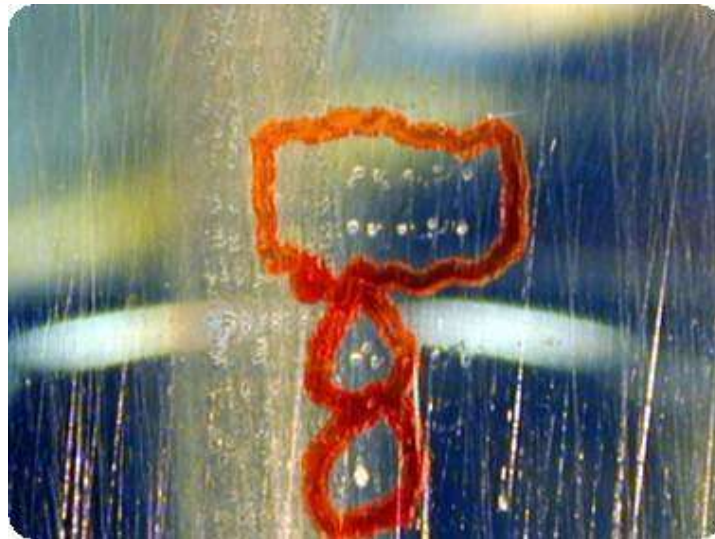


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5.4.3 current leakage...



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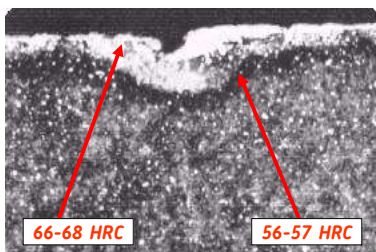
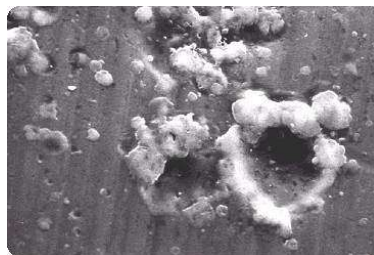
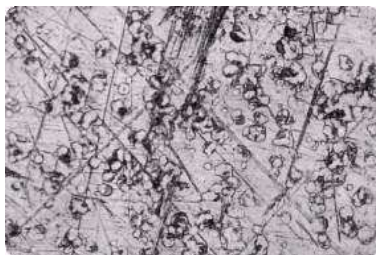
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5.4.3 current leakage...



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5.4.3 current leakage...



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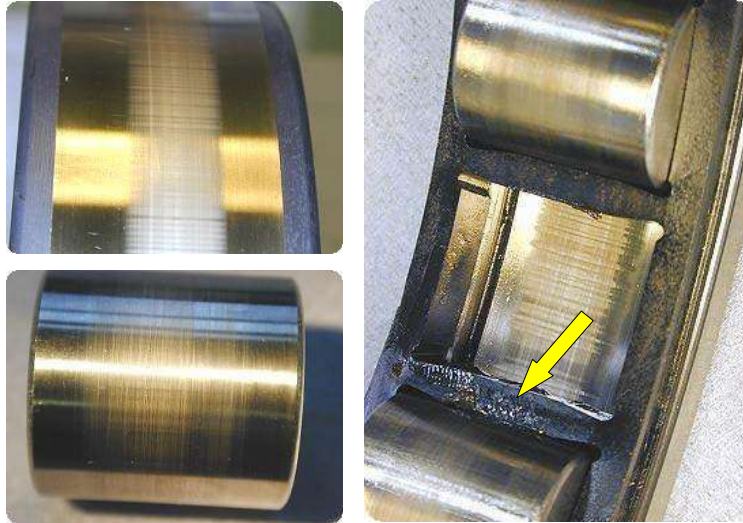
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5.4.3 current leakage...



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5.4.3 current leakage

5.4 Electrical erosion

5.4.3 Current leakage

How to avoid?

- *Symmetric cabling*
- *Proper alignment of stator and rotor*
- *Use insulated bearings*
 - *INSOCOAT*
 - *Hybrid bearings*

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5.5

Plastic deformation

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5.5.1 general definition

- *Permanent deformation occurring whenever the yield strength of the material is exceeded*
- *Typically, this can occur in two different ways:*
 - *On a macro scale, where the contact load between a rolling element and the raceway causes yielding over a substantial portion of the contact footprint*
 - *On a micro scale, where a foreign object is over-rolled between a rolling element and the raceway, and yielding occurs over only a small part of the contact footprint*

Plastic deformation can be divided into:

5.5.2 Overload

5.5.3 Indentation from debris

5.5.4 Indentation by handling

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5.5.2

Overload

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5.5.2 overload...

5.1 Fatigue

5.2 Wear

5.3 Corrosion

5.4 Electrical
erosion

5.5 Plastic
deformation

5.6 Fracture and
cracking

- Static or shock loads
- Plastic deformations
- Depressions at rolling element distance
- Inappropriate handling

5.5.2 Overload

5.5.3 Indentation from debris

5.5.4 Indentation by handling



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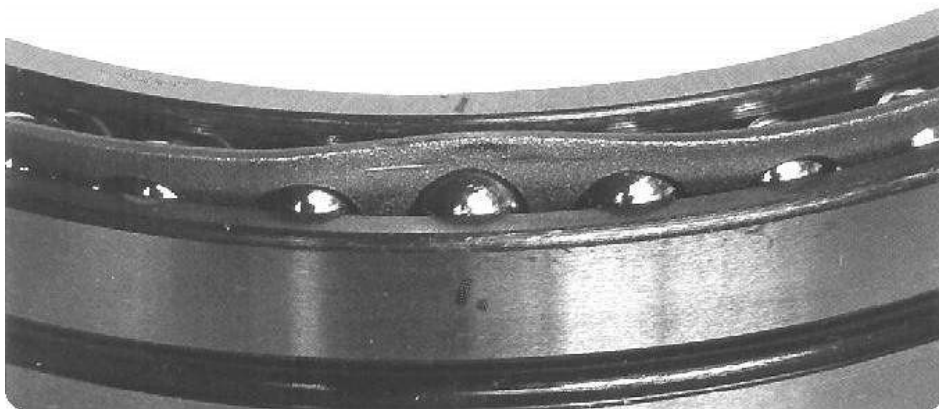
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5.5.2 overload...

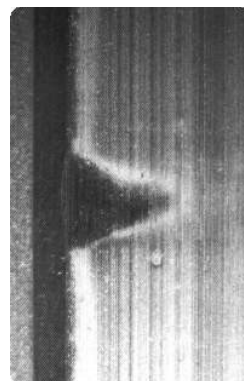
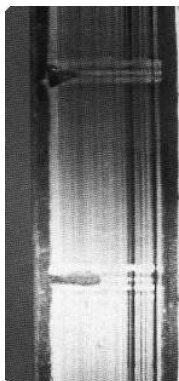
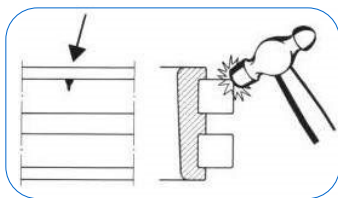


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5.5.2 overload...



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5.5.2 overload...

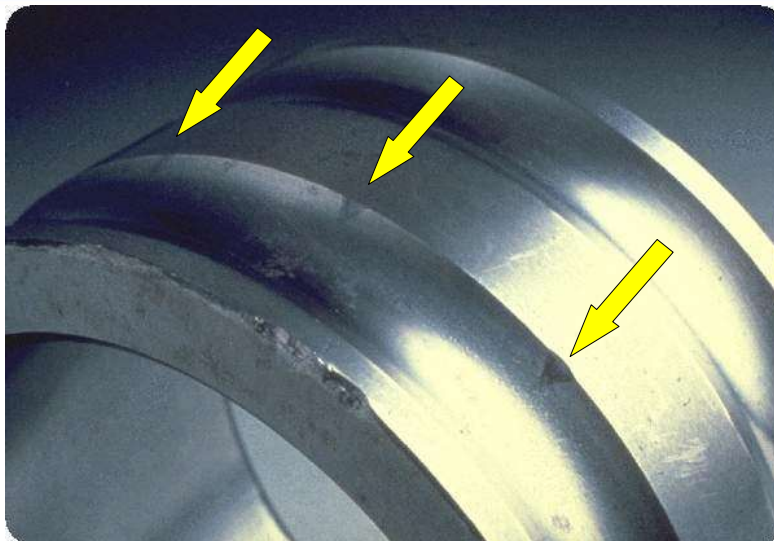


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5.5.2 overload...



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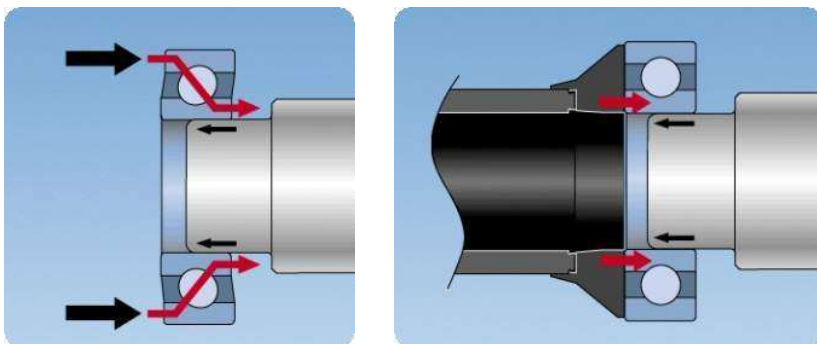
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5.5.2 overload...



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5.5.2 overload...



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5.5.2 overload

5.5 Plastic deformation

5.5.2 Overload

How to avoid?

- *Use appropriate methods for mounting bearings*
- *Use correct tools, use a mounting dolly if appropriate*
- *Follow procedures and instructions carefully*

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5.5.3

Indentation from debris

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5.5.3 indentations from debris...

5.1 Fatigue

5.2 Wear

5.3 Corrosion

5.4 Electrical erosion

5.5 Plastic deformation

5.6 Fracture and cracking

• Localized overloading

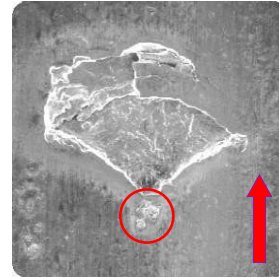
• Over-rolling of particles: dents

• Soft/hardened steel/hard mineral

5.5.2 Overload

5.5.3 Indentation from debris

5.5.4 Indentation by handling

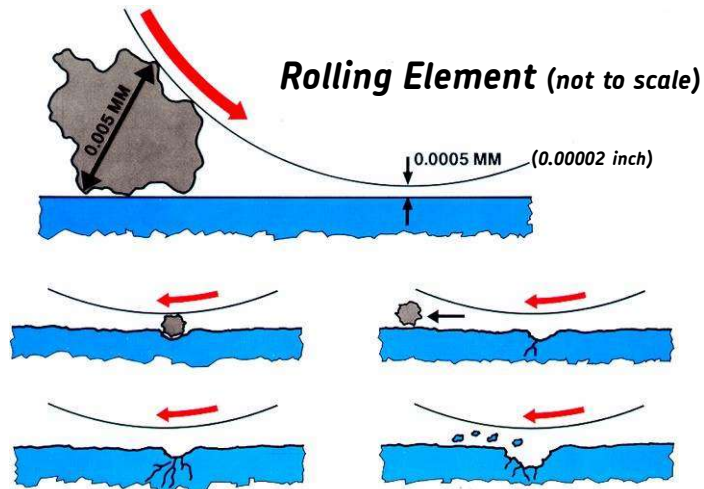


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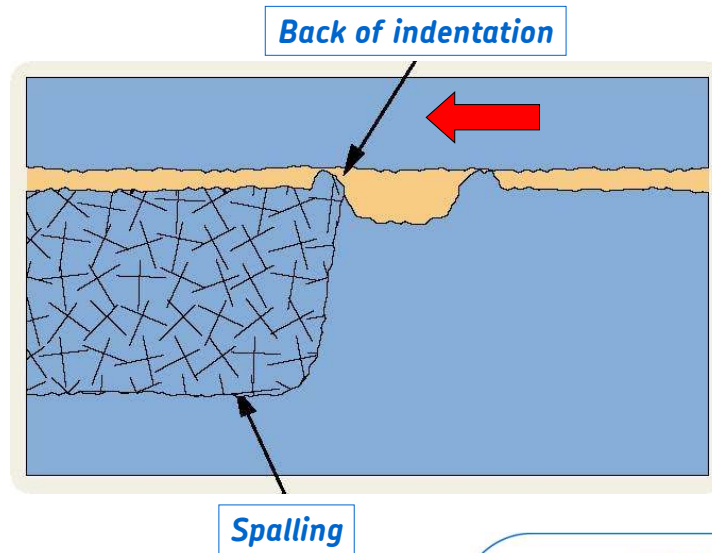
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5.5.3 indentations from debris...

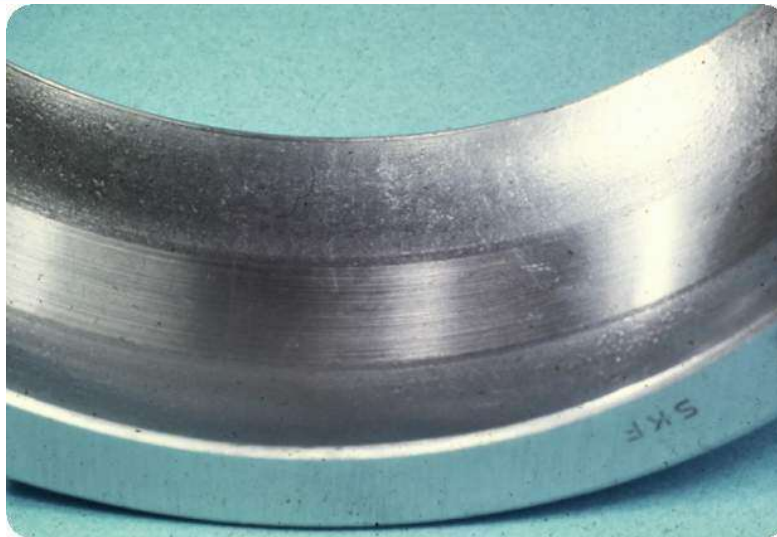


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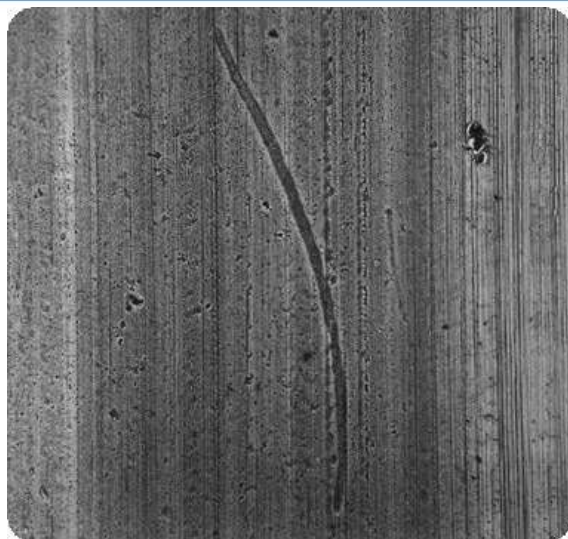
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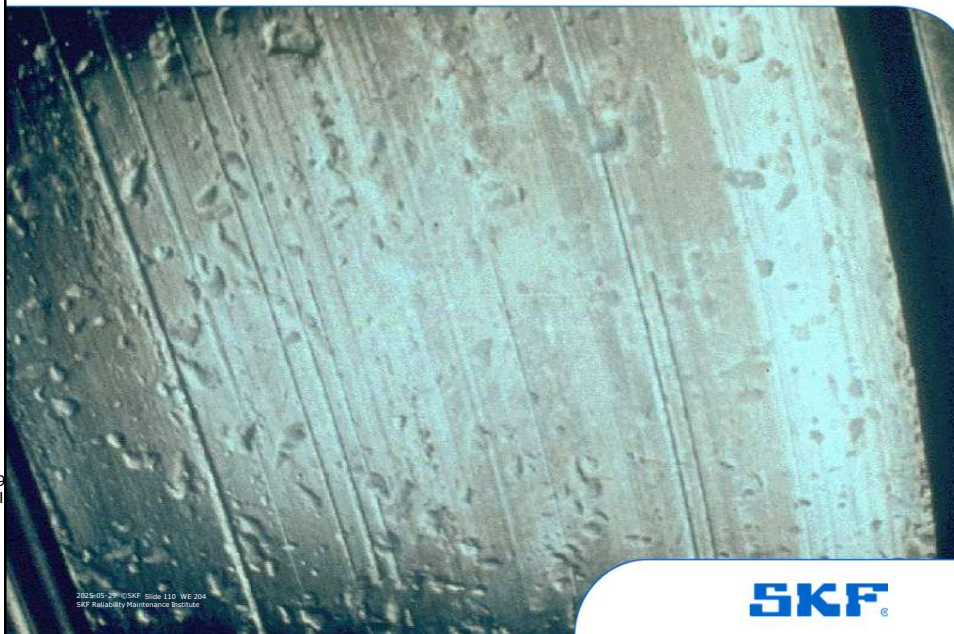


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5.5.3 indentations from debris...



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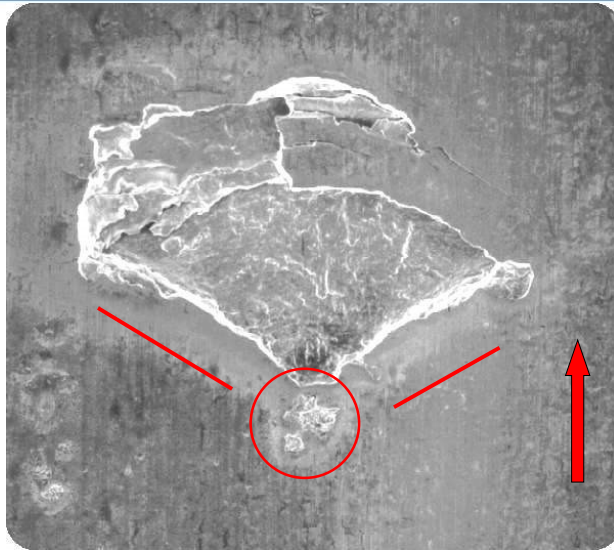
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5.5.3 indentations from debris...



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5.5.3 indentations from debris

5.5 Plastic deformation

5.5.3 Indentation from debris

How to avoid?

- *Use adequate sealing to protect the bearing arrangement*
- *Make sure you operate with appropriate lubrication*
 - *The right lubricant (must be clean too)*
 - *The right quantity*
 - *At the right time (in dirty environment, continuous lubrication)*
- *Make sure to work in clean locations*

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5.5.4

Indentation by handling

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5.5.4 indentation by handling...

5.1 Fatigue

• Localized overloading

5.2 Wear

• Nicks caused by hard/sharp objects

5.3 Corrosion

5.4 Electrical
erosion

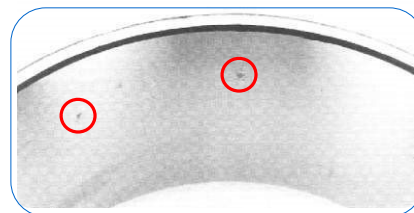
5.5 Plastic
deformation

5.5.2 Overload

5.5.3 Indentation from debris

5.5.4 Indentation by handling

5.6 Fracture and
cracking



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5.5.4 indentation by handling...



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5.5.4 indentation by handling...



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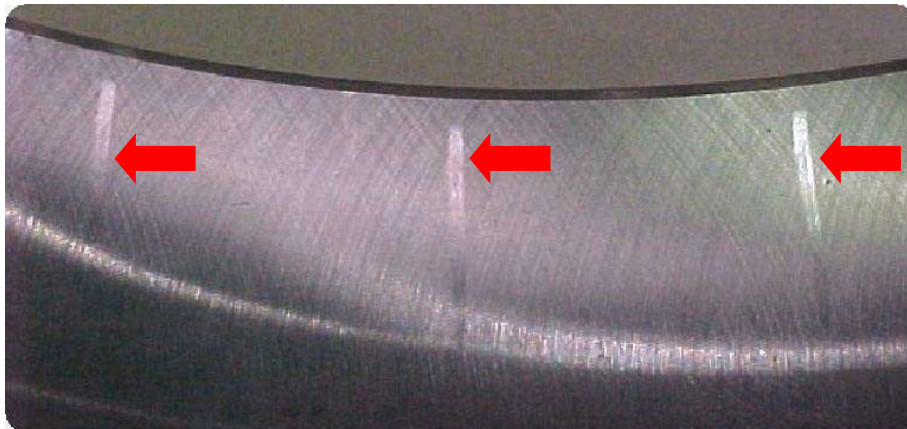
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5.5.4 indentation by handling...



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5.5.4 indentation by handling

5.5 Plastic deformation

5.5.4 Indentation by handling

How to avoid?

- Bearings are “precision” components; treat them very carefully
- Never drop a bearing
- When assembling bearings, make sure components are aligned; use a mounting dolly where needed

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5.6

Fracture and cracking

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5.6.1 general definition

- *Cracking is the incomplete separation of material under stress conditions*
- *Cracks are initiated and propagate when the ultimate tensile strength of the material is exceeded*
- *Fracture is the result of a crack propagating to the point of complete separation of a part of the component*

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5.6.2

Forced fracture

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5.6.2 forced fracture...

5.1 Fatigue

5.2 Wear

5.3 Corrosion

5.4 Electrical
erosion

5.5 Plastic
deformation

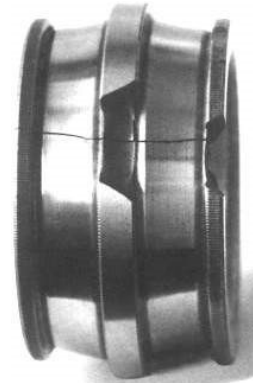
5.6 Fracture and
cracking

- Stress concentration in excess of the material's tensile strength
- Impact/overstressing

5.6.2 Forced fracture

5.6.3 Fatigue fracture

5.6.4 Thermal cracking

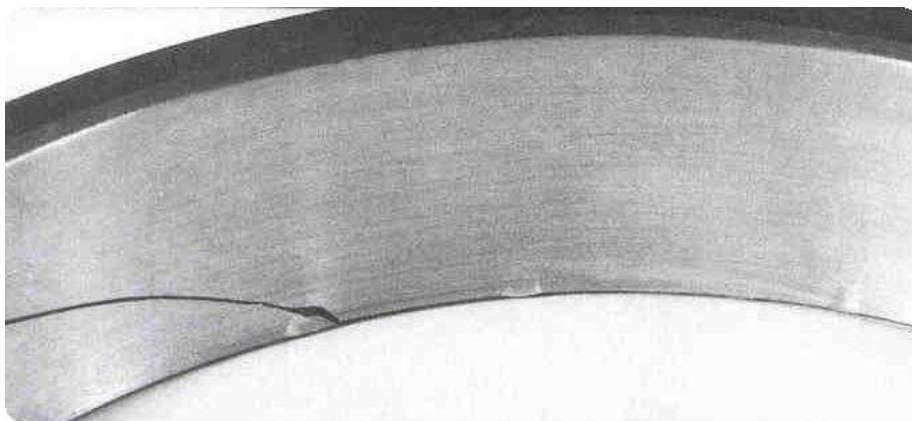


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5.6.2 forced fracture...



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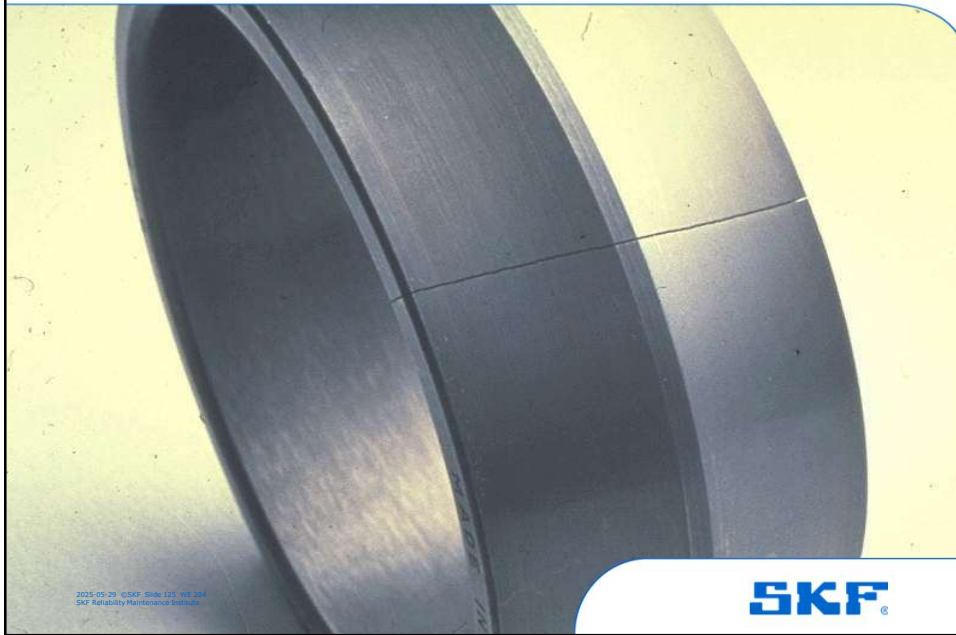
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5.6.2 forced fracture...



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5.6.2 forced fracture

5.6 Fracture and cracking

5.6.2 Forced fracture

How to avoid?

- *Make sure the right fits are used, i.e., not too tight*
- *Use appropriate methods for mounting bearings*
- *Use correct tools, use a mounting dolly if appropriate*
- *Follow procedures and instructions carefully*
- *Never use brute force on bearings*

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5.6.3

Fatigue fracture

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5.6.3 fatigue fracture...

5.1 Fatigue

- Exceeding fatigue strength under bending

5.2 Wear

- A crack is initiated and propagated

5.3 Corrosion

- Causing forced fracture of rings and cages

5.4 Electrical erosion

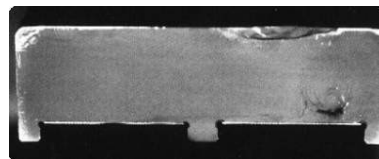
5.5 Plastic deformation

5.6 Fracture and cracking

5.6.2 Forced fracture

5.6.3 Fatigue fracture

5.6.4 Thermal cracking



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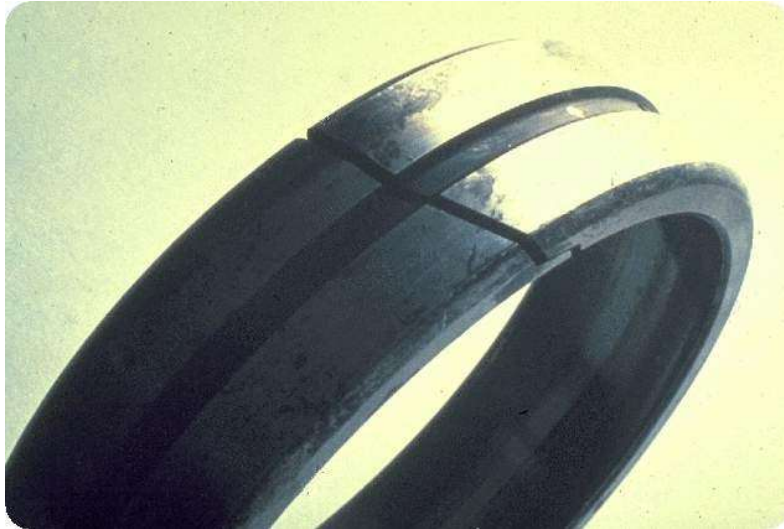
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5.6.3 fatigue fracture...



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5.6.3 fatigue fracture

5.6 Fracture and cracking

5.6.3 Fatigue fracture

How to avoid?

- *Make sure bearing seats have the right form and shape*
- *For split housings, make sure the two halves join correctly*
- *Bearing seats must be clean; no contaminants*
- *Use appropriate tools and methods*

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5.6.4

Thermal cracking

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5.6.4 thermal cracking...

5.1 Fatigue

- Heavy sliding and/or insufficient lubrication

5.2 Wear

- High frictional heat

5.3 Corrosion

- Cracks at right angle to sliding direction

5.4 Electrical
erosion

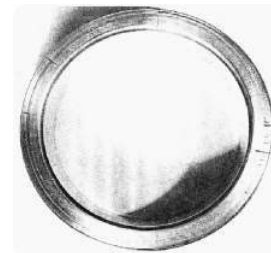
5.5 Plastic
deformation

5.6 Fracture and
cracking

5.6.2 Forced fracture

5.6.3 Fatigue fracture

5.6.4 Thermal cracking



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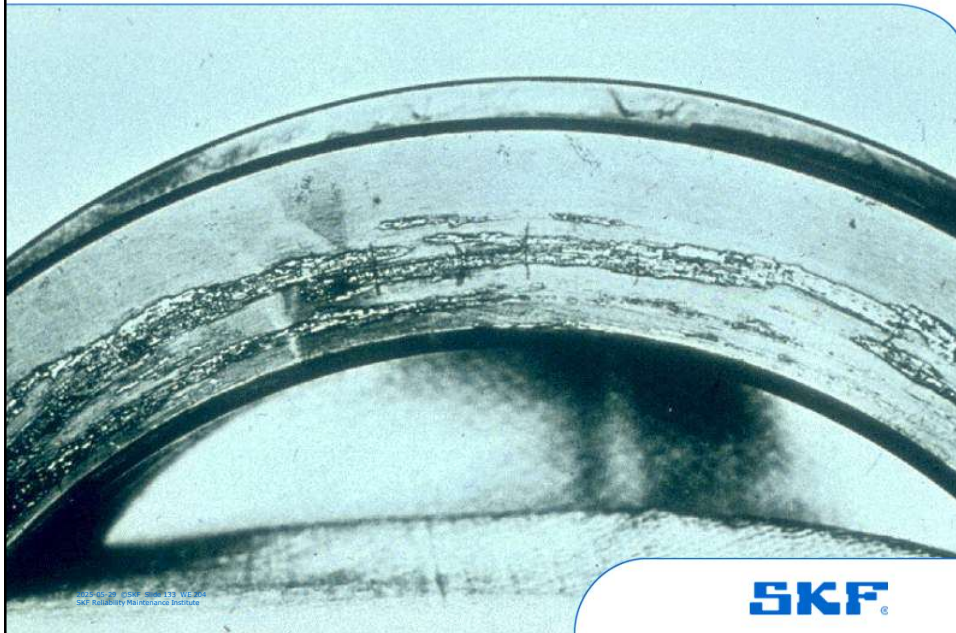
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5.6.4 thermal cracking...



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5.6.4 thermal cracking

5.6 Fracture and cracking

5.6.4 Thermal cracking

How to avoid?

- *Lubricate sliding surfaces, if possible*
- *Adequate lubricant*
- *If sliding is unavoidable, consult SKF; some specific bearings exist*

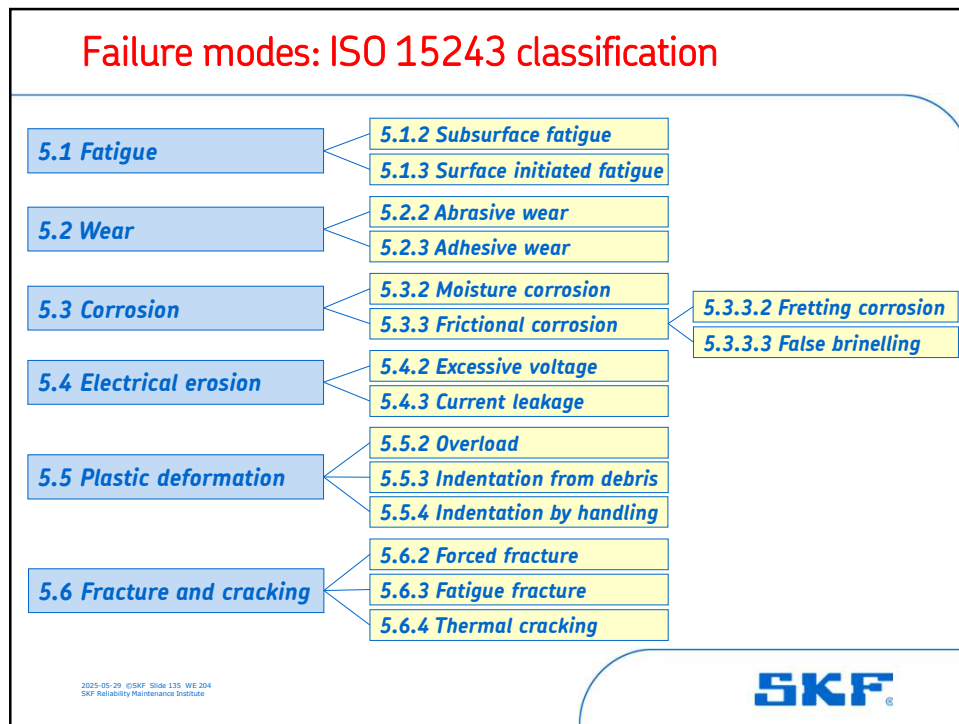
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ISO failure modes versus possible causes

Possible causes		Operating conditions					Environmental factors		Lubrication factors		Mounting factors					Other factors												
		Overload	Over speed	Excessive freq. of load/speed changes	Vibrations	Shaft / housing deflection	Temperature too high / low	dust and dirt ingress	Water ingress	Electrical leakage	Wrong viscosity	(Consistency) additives selection	Lack of lubricant	Excess of lubricant	Impurities	Incorrect handling (shock loads)	Mounting procedures	Fit too tight	Fit too loose	Tilting / misalignment	Incorrect setting	Incorrect locating (clamping)	Storage	Transportation (vibration / shock)	Bearing selection	Equipment design	Manufacturing concerns	Material concerns
Failure modes with characteristics	Fatigue	Flaking, spalling, peeling	++				++	++	++	+	++	++	++	++	++	+	+						++	++	++	++	++	++
		Burnishing, microcracks	++	++	++		++	++	++	+	++	++	++	++	++	+	+						++	++	++	++	++	++
Wear	Abrasive	Excessive wear	++	++	++		++	++	+	++	++	++	++	++	++		+	++	+	++				++	++	++	++	++
		Scratches, scores	++	++			++				++				++	++	++						++					
	Adhesive	Seizing marks, smearing	++	++		++					++	++	++	++	++		+	++	+					++	++	++	++	++
		Hot runners	++	++	+	++					++	++	++	++	++		++	++	+	++	++			++	++	++	++	++
Corrosion		Moisture corrosion					-		++		++				+	+							++	++		+	++	++
		Fretting corrosion	++	++	++	+													++	+	++	++		++	++	++	++	++
		False brinelling			+	++						+											+	++	++		++	++
Electrical erosion		Craters, fluting								++																++		
Plastic deformation		Depressions	++				++	++						+		++	++	++	++	++	++	++	+	+	++	++	++	++
		Debris indentation						++								++	++	++								++	++	++
		Nicks, gouges															++	++								++		
Fracture		Forced fracture	++	++			++									++	++	++		++	++	++		++	++	++	++	++
		Fatigue fracture	++	++	+	++												+	+	+	+			++	++	++	++	++
		Thermal cracking	++	++	+	++	++				+	+	+				++								+	++	++	++

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A damaged bearing

- *The bearing is most often a victim*
- *Something did not work properly*
- *The bearing needs to be examined to find out about the root cause of the problem*
- *But the bearing is not the only part to be looked at*
- *There is more!*

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A damaged bearing – the lubricant?



Water in the grease?

*Used: yellow
colour
Fresh: brown colour*



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A damaged bearing – the sealing?



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Last word



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How to become an experienced analyzer?

Practice

Practice

And last but not least:

Practice!

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How to become an experienced analyzer?

Practical & Reporting

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Securing Evidence

- Collect Operating Data, Monitoring Data
- Collect Lubricant Samples
- Check Bearing Environment(s)
- Assess Bearing(s) in Mounted Condition
- Mark Mounting Position(s)
- Remove, Mark and Bag Bearing(s) and Parts
- Check Bearing Seats

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Conducting the Analysis

- Examine Bearing(s) and Parts
- Record visual observations
- Use the Failure Modes to eliminate improbable possible causes and determine the original cause of the failure.
- Contact external resources for assistance, if needed
- Initiate corrective action, if desired.

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Brand Protection

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**DON'T JEOPARDIZE YOUR BUSINESS
BUY GENUINE SKF PRODUCTS**

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Verification tools



[SKF Authenticate app](#)

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Download on the
App Store



Talk to an expert?
genuine@skf.com

**Verification
reply within 24
hours**

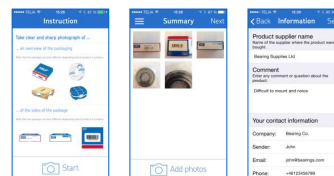
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SKF Authenticate App

- Let SKF experts verify your products



- Counterfeits products exist in all markets
- Are extremely difficult to distinguish from genuine
- Use SKF Authenticate App to check your products

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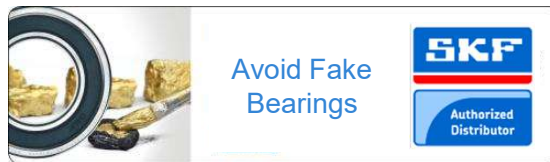
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SKF Authenticate App

- Don't take any risks, verify products before installation

Through the SKF Authenticate App you can:

- In a convenient way, take and send photos of products/packages to SKF
- Have the items verified by SKF experts, free of charge
- Minimize risk of getting counterfeits into your operations
- Identify suppliers of counterfeit, and remove them from your supply chain



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genuine@skf.com sample reply

RE: 230804_Lim-Precision Locknut



Arjun R Vishwakarma on behalf of genu
To Lim Ann Hock



8/4/2023

Dear Lim,

Based on the available photos sent we have concluded that the product shown on the pictures is **counterfeited, i.e. not genuine.**

Verified item is:

[KMT8 with black nickel treatment]

Regarding the other product shown on the picture "KMT 8 Gold", our investigation concludes that there is **nothing suspicious** about the product.

Please provide SKF with supplier information (e.g. invoice, receipt or similar) for these items, please do so by replying to this email and attach relevant documents. This information is important to enable SKF to take actions against this illicit trade and remove counterfeits from the market.

Best regards,

Arjun Vishwakarma
SKF Group Brand Protection

General inquiries: genuine@skf.com

SKF Group Headquarters
Chassis Division, P.O. Box 330, 2200 SB, The Netherlands

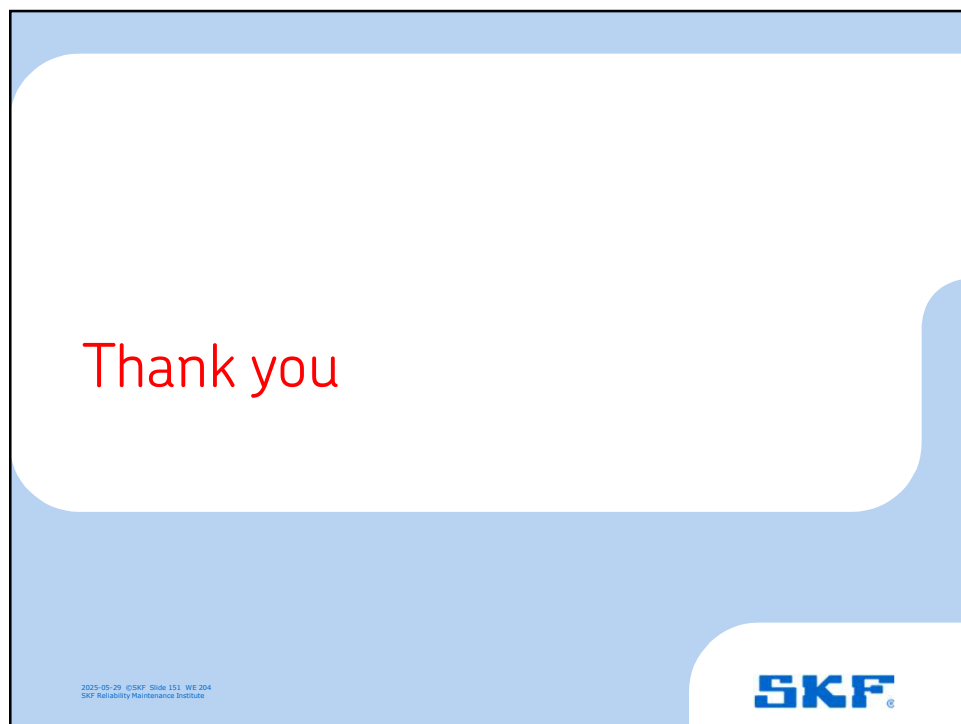
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